



SUNFLASER

OPTOELECTRONIC SOLUTIONS

SFA60

Laser Rangefinder Module



Product Manual

Version 202506V1

Disclaimer

This product manual is intended to provide reference specifications and performance parameters of the product. If certain specifications are not included in this document, please contact SunFlaser Technology (Chengdu) Co., Ltd. directly for further confirmation. All contents of this manual and the specifications of the ranging module are subject to change without prior notice.

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Safety Instructions

For the safe use of the ranging module, please carefully follow the safety instructions provided in this manual. Do not attempt to disassemble, modify, or repair any part of the ranging module. Do not tamper with or attempt to alter the performance of the ranging module.

Intended Use

- Take precautions against electrostatic discharge (ESD). Do not touch any electronic components or parts of the ranging module without proper protective measures.
- Keep this manual readily accessible to all personnel operating the ranging module.
- Operate the ranging module only within the specified voltage and power range.
- Operate the ranging module in a clean, dry, and ESD-protected environment.
- Do not touch the optical glass lens with your fingers.
- Avoid exposing the ranging module to environments with drastic temperature fluctuations.

Improper Use

- Using the ranging module with the host system before verifying that all functions are operating properly;
- Operating the ranging module without first reading and understanding this manual and all safety instructions;
- Disassembling, modifying, or altering the ranging module without authorization from SunFlaser Technology (Chengdu) Co., Ltd.;
- Sending operating commands other than those specified in this manual;
- Using or testing the ranging module in non-standard, abnormal, or unsuitable environments;

- › Operating the ranging module in a manner inconsistent with the instructions provided in this manual;
- › Continuing to operate the ranging module after improper handling or misuse;
- › Using a ranging module with obvious faults, defects, or damage;
- › Performing test operations within the blind zone of the ranging module.

Improper use may result in the following risks:

- › Eye injury
- › Incorrect results
- › Measurement errors
- › Property damage
- › Equipment malfunction

Operator Responsibilities

- › Be familiar with the basic knowledge required for the correct operation of the ranging module;
- › Understand and comply with the operation and maintenance requirements of the ranging module;
- › Prevent damage to the ranging module during storage and transportation;
- › Keep the ranging module clean and well maintained;
- › Ensure that the ranging module is stored and operated in a safe environment;
- › Do not expose the ranging module to strong mechanical shock or impact;
- › Do not expose the ranging module to environments with drastic temperature fluctuations.

Avoid Measurement Errors

- › Pay attention to all factors that may affect the performance of the ranging module;

- After the Laser Rangefinder Module has been subjected to improper handling (such as vibration, impact, or dropping), or before performing critical measurement tasks, it is recommended to verify the performance specifications of the ranging module.

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01

Product Overview

SFA60 Laser Rangefinding Module adopts a 650nm laser device and is developed based on phase-shift ranging technology.

The module consists of a laser transmitting unit, receiving optical assembly, and main control circuit. It is equipped with a UART (TTL_3.3V) communication interface and provides standard communication protocols and debugging solutions, enabling rapid integration and secondary development for users.

Featuring a compact size, lower power consumption, high measurement accuracy, and reliable performance, the product complies with Class II eye safety standards. It can be widely applied in smart devices, industrial inspection, compact rangefinding instruments, and other related applications.

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Key Features

- Rangefinding Function: Supports measurement range from 0.03m to 60m with 1mm resolution, delivering high-precision and stable output

- Measurement Modes: Supports both single measurement and continuous measurement modes
- Communication Control: UART (TTL_3.3V) serial command control with support for multiple baud rates
- Parameter Configuration: Supports module address setting, measurement offset calibration, and laser enable/disable control
- Status Monitoring: Real-time access to voltage, firmware version, serial number, and fault status codes
- Multi-Module Networking: Supports one-master multiple-slave architecture, allowing up to 128 modules to operate collaboratively

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Key Specifications

Item		Parameter
Performance Specifications	Eye Safety Class	Class II (IEC 60825-1)
	Laser Power	< 1 mW
	Laser Wavelength	650 ± 40 nm
	Spot Diameter	< 8 mm @ 10 m < 20 mm @ 20 m < 40 mm @ 40 m
	Maximum Measuring Range	60 m @ 100% reflectivity, 1kLux ambient light interference
	Minimum Measuring Range	0.03 m
	Measurement Accuracy	±(2 mm + D × (1 / 10000)) @ 20 %–100 % reflectivity, 1kLux ambient light interference
	Maximum Measurement Frequency ^①	4 Hz (30 Hz optional) @ 100 % reflectivity, 1kLux ambient light interference, 0.03–10 m
	Distance Resolution	1 mm
Electrical Specifications	Interface Type	UART (TTL_3.3V)
	Supply Voltage	DC 3.5–4.2 V; recommended 4.0 V
	Standby Power Consumption	< 60 mW
	Operating Power Consumption	< 320 mW
	Startup Time	100 ms
Physical Characteristics	Weight	2 g ± 0.5 g
	Dimensions	16.5 mm × 17 mm × 7 mm (L × W × H)
Environmental Specifications	Operating Temperature	-10 - 50 °C (14 - 122 °F)
	Storage Temperature	-25 - 60 °C (-13 - 140 °F)

Note:

The specified performance is based on line-of-sight conditions with visibility not less than 20 km, relative humidity $\leq 80\%$, and target reflectivity ≥ 0.3 .

① Low target reflectivity or longer measuring distances may reduce the measurement frequency.

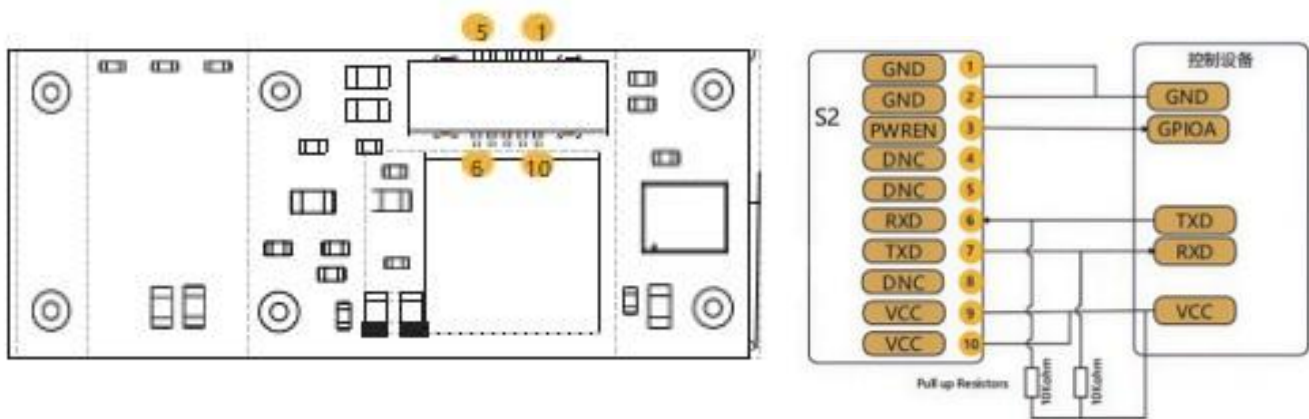
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Interface and Connection Description

Interface Type: UART (TTL_3.3V)

Baud Rate: 115200bps (default)

The interface definitions are detailed in the table below:



Module Pin Layout and Recommendations:

Pin No.	Name	Function	Default Level	Description
Pin 1	GND	Ground	Determined by ground connection	Module power ground
Pin 2	GND	Ground	Determined by ground connection	Module power ground
Pin 3	PWREN	Digital Input	Low	Module power enable pin, triggered by high level; VIH=2.0 V, VIL=0.5 V
Pin 4	DNC	None	High	Leave unconnected
Pin 5	DNC	None	High	Leave unconnected
Pin 6	RXD	Digital Input	High	Module UART receive pin
Pin 7	TXD	Digital Output	High	Module UART transmit pin, push-pull output by default
Pin 8	DNC	None	High	Leave unconnected
Pin 9	VCC	Power Input	Determined by power supply	Typical module power requirement: 300mA @ DC 3.5V
Pin 10	VCC	Power Input	Determined by power supply	Typical module power requirement: 300mA @ DC 3.5V

Note:

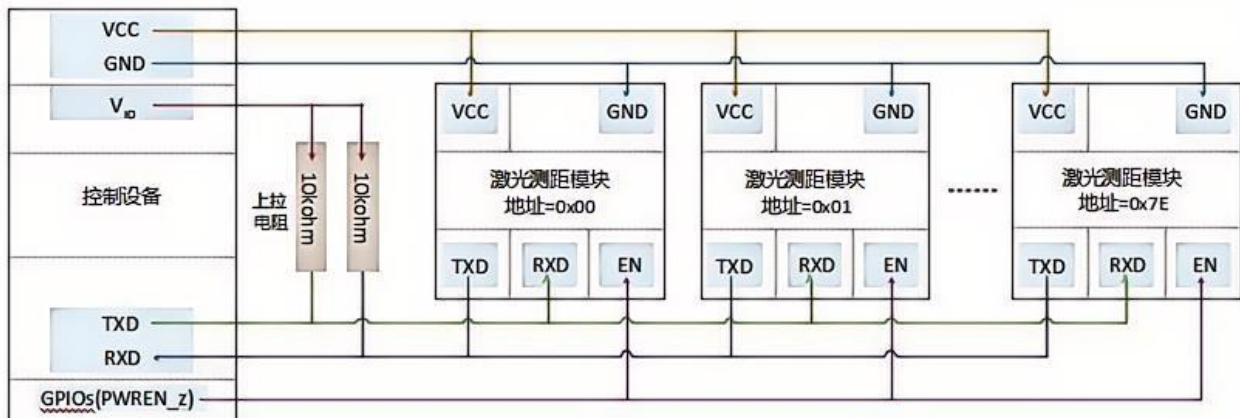
When the PWREN input voltage is higher than the VIH threshold, PWREN is recognized as a logic high level.

When the PWREN input voltage is lower than the VIL threshold, PWREN is recognized as a logic low level.

To ensure reliable module power-on and power-off control, please pay attention to the initial level and control level of the PWREN pin.

Parameter	Minimum Value	Maximum Value
PWREN	-0.3V	5.5V
TXD	-0.3V	VCC + 0.3V
RXD	-0.3V	VCC + 0.3V
VCC	-0.3V	5.5V
GND	0V	0V

Multi-Module Connection:



Note:

The number of modules within a single network segment should not exceed 8.

The maximum number of modules across multiple network segments must not exceed 128.

In addition, the pull-up resistor value on the bus should be adjusted appropriately according to the number of connected modules. When multiple modules are connected, the pull-up resistor value may be reduced accordingly.

Connector Selection:

FPC Connection

Users may select a compatible FPC cable and connect it to the corresponding FPC interface of the module, as shown in the figure below.



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Module Connection and Communication

For detailed information, please refer to Appendix 1:

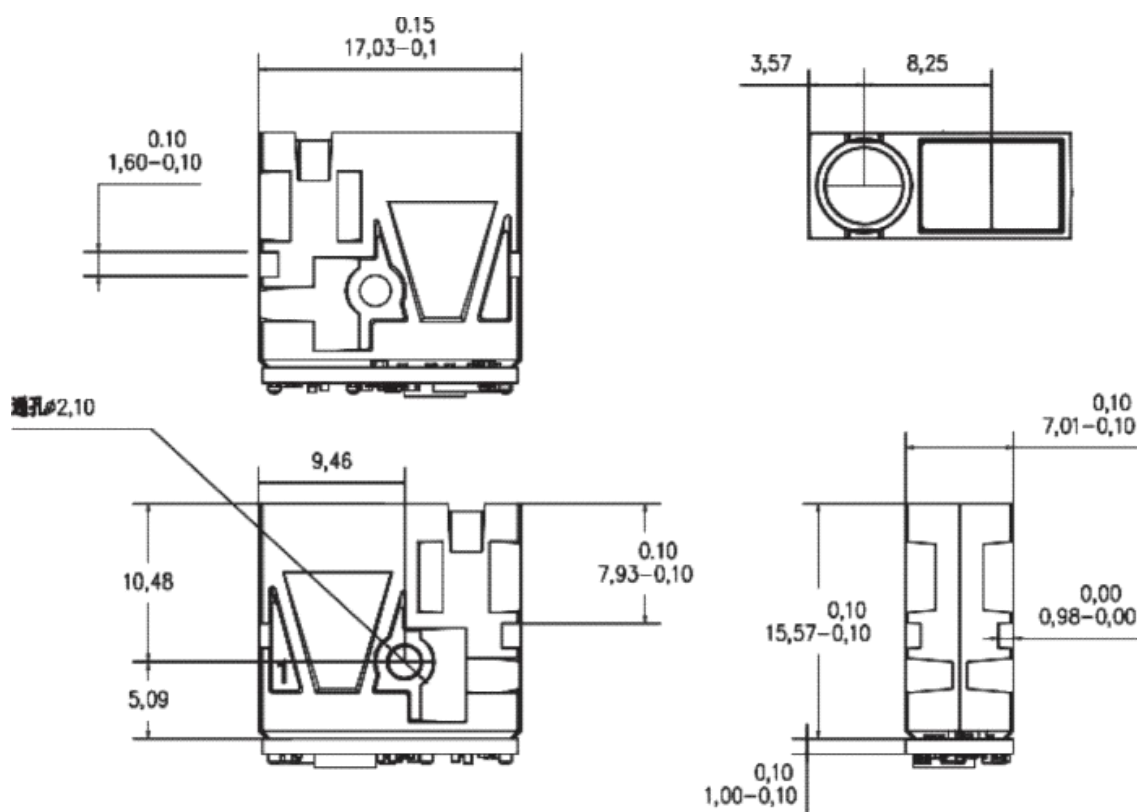
1. Communication Interface
2. Control Flowchart
3. Measurement Modes: Single Measurement / Continuous Measurement
4. Data Packet Format
5. Control Registers
6. Control Commands
7. Status Codes
8. Command Reference List

For the target reflectivity parameter table, please refer to Appendix 2.

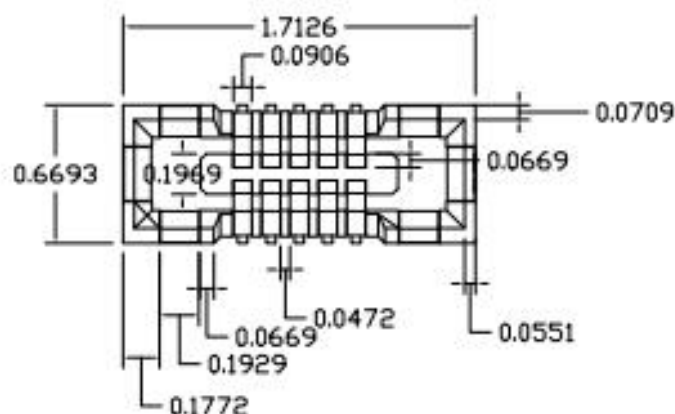
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Mechanical Structure and Mounting Dimensions

The module dimensions are shown below:



FPC Connector Mechanical Dimensions:



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Precautions for Use

- **Do not look directly into the laser emission aperture to avoid eye injury (complies with Class II laser safety standards).**
- **Do not operate the product beyond the specified electrical parameters.**
- **Do not perform wiring operations while the product is powered on.**
- **Please follow the wiring instructions in this manual strictly.**
- **Keep the front surface of the optical lens clean to ensure proper product performance.**

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Safety Precautions

- To ensure the safety of both operators and test subjects, the following safety measures have been incorporated into the design of the SFA60 Laser Rangefinder Module;
- Safety design and analysis are conducted in accordance with GJB 900A-2012, General Requirements for Equipment Safety;
- Non-flammable materials are used, and all mechanical and electrical interfaces are securely connected;
- Appropriate design measures are implemented to prevent moisture accumulation that could result in short circuits;
- The module operates below the safety voltage limit for the human body.

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Maintenance Instructions

Packing List

The standard items supplied with the product are listed in the table below. Optional accessories may not be included in this list and shall be subject to the actual product delivered.

No.	Item	Quantity	Remarks
1	Laser Rangefinder Module	1 unit	-
2	Certificate of Conformity	1 copy	-
3	Packaging Box	1 pc	-
4	FPC Cable	1 pc	-

Module Cleaning

Cleaning of Optical Components

- Dust particles should be removed using an air blower;
- Fingerprints should first be cleaned with absorbent cotton lightly moistened with a small amount of alcohol-ether mixture, and then wiped with a clean lens cloth.

Cleaning of Mechanical and Electronic Components

- With the power disconnected, gently wipe mechanical and electronic components with alcohol and allow them to dry naturally before use;
- Keep the ranging module, connectors, and cables away from moisture and contaminants whenever possible;
- Ensure that the equipment is completely dry before packaging.

Inspection and Maintenance

Routine Maintenance

When the product is used for the first time, or after replacement of related modules or components, both visual inspection and power-on inspection shall be performed. For products under normal operation, only a power-on inspection is required before use.

- Visual Inspection Procedure
 - Check whether the product appearance is normal;

- Check whether the cable connections are correct and securely connected.
- Power-On Inspection Procedure
 - Perform the power-on procedure and verify that the standby current is correct;
 - Start the self-test function of the product;
 - After the inspection is completed, perform the power-off procedure.

Periodic Maintenance

Under normal operating conditions, the Laser Rangefinder Module does not require routine maintenance. If the module has been stored in a dust-free environment for more than one year, maintenance should be performed, including general inspection and power-on inspection.

- General Inspection (Performed with Power Off)
 - All markings and serial numbers on the product and test cable connectors shall be correct and clearly legible;
 - All screws on the panel shall be securely fastened;
 - The optical glass of the product shall be free from visible defects that may affect normal operation, such as spots, pitting, water stains, fungus, fingerprints, dust particles, or cracks.
- Comprehensive Power-On Inspection and Maintenance
 - Connect the product power supply in sequence;
 - Perform the power-on procedure and verify that the standby current is correct;
 - Start the product self-test and complete the self-test procedure;
 - Perform the power-off procedure after completion of the inspection.

Troubleshooting

The following only lists some possible fault phenomena that are not caused by the product itself.

If other technical issues occur during product use, please contact our company for support.

Troubleshooting

- The computer does not detect the serial port device after connecting to the USB-to-TTL module:
 - Check whether the CH340 driver is installed on the computer. If not, the “CH340 Driver” application can be found and installed from the Liangxin documentation package.
 - Check whether the communication connection between the USB-to-TTL module and the computer is stable.
 - Check whether the computer USB port is damaged or whether the USB-to-TTL module is faulty.
- The module powers on successfully and connects to the computer, but does not respond to commands:
 - Check whether the module TXD and RXD connections are reversed.
 - Check whether the baud rate is configured correctly.
 - Check whether the module TXD and RXD are configured as open-drain outputs. If operating in open-drain mode, verify that appropriate pull-up or pull-down resistors are connected.
- The module communicates normally, but the received data appears garbled:
 - Check whether the serial debugging tool is configured for HEX display (hexadecimal display).
 - Check communication stability:
 - ① Check for nearby electromagnetic interference or other signal interference sources. If present, move the device away from the interference source as much as possible.
 - ② Check whether the serial communication cable is excessively long. For standard UART (TTL-level) communication, the cable length is recommended to be within 1 meter.

Packaging and Transportation Requirements

› **Packaging**

After unpacking, if the product needs to be placed back into storage, it shall be repackaged using the original packaging whenever possible.

If the product needs to be returned to the manufacturer, the original packaging is recommended. If alternative packaging is used, it shall not cause any degradation in product performance or physical damage.

› **Transportation**

Repackaged products may be transported by automobile, railway, aircraft, or ship. During transportation, the package shall be securely fixed to the transport vehicle to avoid impact, improper handling, and exposure to rain or snow.

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Technical Support and Service

If a product malfunction occurs and factory return is required for fault analysis, troubleshooting, or repair, SunFlaser Technology (Chengdu) Co., Ltd. provides a one-year warranty and lifetime technical support from the date of product delivery.

During the warranty period, product failures caused by manufacturing or quality issues will be repaired or replaced free of charge by SunFlaser. For product damage caused by improper operation or other user-related factors, repair and replacement costs will be charged according to the actual situation.

This product is a precision optical instrument. Please handle and operate it carefully during use. If you have any additional questions regarding operation or maintenance, please contact our after-sales support personnel at any time.

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Contact Information

Website: www.sunflaser.com

Tel: +86 186 2827 7862 (Sales Manager: Mr. Cao)

Address: No. 58, Zhenxing West 1st Road, Building 1, Room 201, Jinniu District, Chengdu, Sichuan, China

Appendix 1:

Module Communication

1 UART Interface

Modules supporting auto-baud rate detection: 9600bps, 19200bps, 38400bps, 115200bps;

Fixed baud rate modules: 115200bps (baud rates available: 4800bps, 9600bps, 38400bps, 115200bps, 19200bps).

Format: 1 start bit, 8 data bits, 1 stop bit, no parity, no flow control.

For modules supporting auto-baud rate, if the module does not

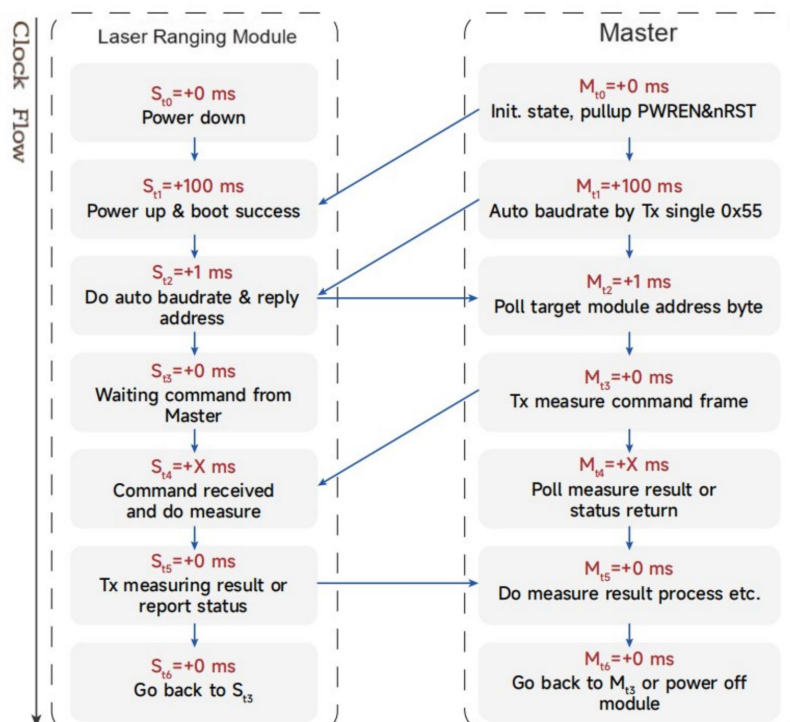
receive the auto-baud rate handshake byte 0x55 within 2.5 seconds after power-up, or receives an incorrect handshake byte, it will communicate at the fixed baud rate of 115200bps.

2 Control Flowchart

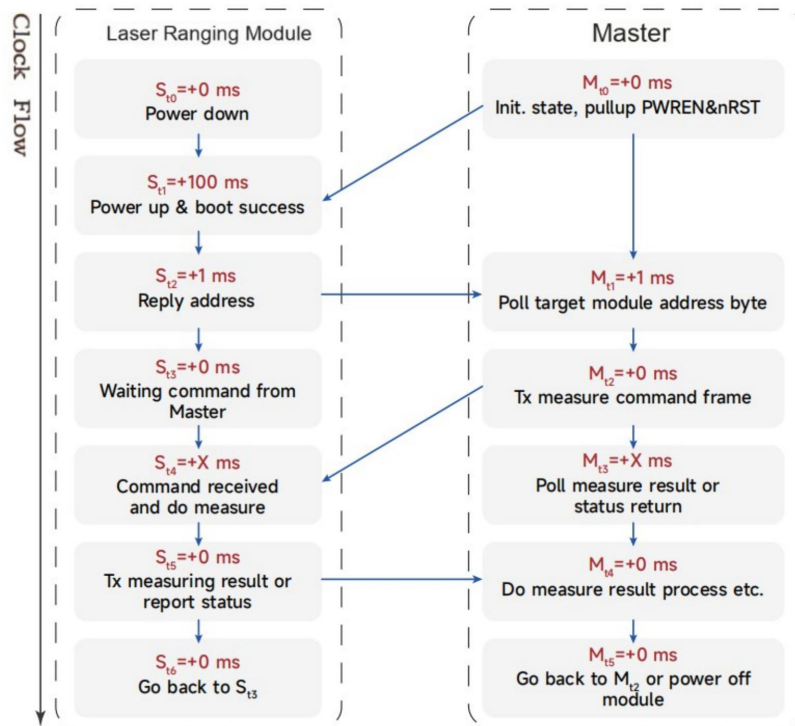
All communication commands are initiated by the host, and the laser ranging module acts as a slave responding to the host's commands.

The communication timeline is shown in the figures.

Auto-baud rate module:



Fixed baud rate module:



In the initial state, the slave (laser ranging module) is in power-down mode before the host pulls its PWREN pin high. After the PWREN pin is pulled high, if the slave supports auto-baud rate, it will enter the auto-baud rate handshake phase after approximately 100ms of self-check. The host then sends the handshake byte (0x55) to the slave at the desired baud rate. If communication is successful, the slave replies with 1 byte of data, which is the slave's communication address (1 byte). In a one-master-multiple-slave scenario, the host may receive address replies from multiple slaves

after sending the handshake byte, which could cause a UART bus conflict. This byte should be ignored.

After successful auto-baud rate setting, communication between master and slave is established.

3 Measurement Modes

3.1 Single Measurement

Send a single measurement command once. Upon successful measurement, one measurement result is returned.

3.2 Continuous Measurement

Send a continuous measurement command once. The module remains in continuous measurement mode, returning a result for each successful measurement. To exit continuous measurement, the host must send a 1-byte command 0x58 (ASCII character 'X') during a measurement interval.

4 Data Packet Format

Head	RW	Address	Register	Payload count	Payload	Checksum
8 bits	1 bit	7 bits	16 bits	16 bits	Length*16 bits	8 bits
Byte[0]	Byte[1]		Byte[2:3]	Byte[4:5]	Byte[6:N]	Byte[N+1]
Checksum= (Byte[1]+Byte[2]+Byte[3]+ . . . +Byte[N])&0xFF						
R/W (Read/Write Direction) : Master writing data to slave——0; Master reading data from slave——1;						

Address: The address is only 7 bits, range: 0x00 ~ 0x7F. 0x00 is the factory default slave address. 0x7F is the broadcast address from one master to multiple slaves.

5 Control Registers

The module contains numerous internal registers. Users can control the module by accessing these registers.

No.	Register	Name	Function
1	0X0000	REG_ERR_CODE	System Status Code
2	0X0006	REG_BAT_VLTG	Input Voltage
3	0X0010	REG_ADDRESS	Module Address
4	0X0012	REG_OFFSET	Result Offset
5	0X0020	REG_MEA_START	Initiate Measurement
6	0X0022	REG_MEA_RESULT	Measurement Result
7	0X01BE	REG_CTRL_LD	Laser Diode Control

6 Control Commands

6.1 Read Laser Module Status

Send: 0xAA 0x80 0x00 0x00 0x80

Receive: 0xAA 0x80 0x00 0x00 0x00 0x01 0xYY 0xZZ Check

Module Status: 0xYYZZ

Slave Address: 0x00 Register Address: 0x0000

Description: After the previous command is completed, the host sends this command to read the module's current status code.

The status code indicates any existing software or hardware errors.

If no issues exist, the status code is 0x0000.

6.2 Read Hardware Version

Send: 0xAA 0x80 0x00 0x0A 0x8A

Receive: 0xAA 0x80 0x00 0x0A 0x00 0x01 0xVV 0xYY Check

Hardware Version: 0xVVYY

Slave Address: 0x00 Register Address: 0x000A

Description: After the previous command is completed, the host sends this command to read the module's hardware version.

6.3 Read Software Version

Send: 0xAA 0x80 0x00 0x0C 0x8C

Receive: 0xAA 0x80 0x00 0x0C 0x00 0x01 0xVV 0xYY Check

Software Version: 0xVVYY

Slave Address: 0x00 Register Address: 0x000C

Description: After the previous command is completed, the host reads the module's software version.

6.4 Read Module Serial Number

Send: 0xAA 0x80 0x00 0x0E 0x8E

Receive: 0xAA 0x80 0x00 0x0E 0x00 0x02 0xAABBCCDD Check

Module Serial Number: 0xAABBCCDD

Slave Address: 0x00 Register Address: 0x000E

Description: After the previous command is completed, the host reads the module's serial number.

6.5 Read Input Voltage

Send: 0xAA 0x80 0x00 0x06 0x86

Receive: 0xAA 0x80 0x00 0x06 0x00 0x01 0x32 0x19 Check

Input Voltage: 3219mV

Slave Address: 0x00 Register Address: 0x0006

Description: After the previous command is completed, the host reads the module's input voltage (mV, BCD format).

6.6 Read Current Offset

Send: 0xAA 0x80 0x00 0x12 0x92

Receive: 0xAA 0x00 0x00 0x12 0x00 0x01 0xVV 0xYY Check

Module Offset: 0xVYYY

Slave Address: 0x00 Register Address: 0x0012

Description: After the previous command is completed, the host reads the offset value.

6.7 Read Measurement Result

Send: 0xAA 0x80 0x00 0x22 0xA2

Receive: 0xAA 0x00 0x00 0x22 0x00 0x03 0xAABBCCDD 0x0101 Check

Measurement Result: 0xAABBCCDD

Slave Address: 0x00 Register Address: 0x0022

Description: After the previous command is completed, the host reads the measurement result.

6.8 Set Module Address

Send: 0xAA 0x00 0x00 0x10 0x00 0x01 0x00 0xYY Check

Receive: 0xAA 0x00 0x00 0x10 0x00 0x01 0x00 0xYY Check

Slave address set to 0xYY (address uses only bits [6:0]; other bits are ignored).

Note: Do not set the slave address to the broadcast address 0x7F. This address is reserved for one-master-to-multiple-slave networks, requiring all slaves to measure distance simultaneously but only report when polled individually.

Slave Address: 0x00 Register Address: 0x0010

Description: After the previous command is completed, the host sets

the slave address. This address is non-volatile (retained after power-off).

6.9 Set Module Offset

Send: 0xAA 0x00 0x00 0x12 0x00 0x01 0xZZ 0xYY Check

Receive: 0xAA 0x00 0x00 0x12 0x00 0x01 0xZZ 0xYY Check

Slave Address: 0x00 Register Address: 0x0012

Description: After the previous command is completed, the host sets the slave offset. The offset represents the starting point for distance measurement. Factory default is 0. Example: If offset 0xZZYY = 0x007B (+123), the final measured value will be increased by 123mm. If offset 0xZZYY = 0xFF85 (-123), the final measured value will be decreased by 123mm.

6.10 Turn Laser On/Off

Send: 0xAA 0x00 0x01 0xBE 0x00 0x01 0x00 0xZZ Check

Receive: 0xAA 0x00 0x01 0xBE 0x00 0x01 0x00 0xZZ Check

Slave Address: 0x00 Register Address: 0x01BE

Description: After the previous command is completed, turn the laser on or off. Laser ON: 0xZZ = 0x01, Laser OFF: 0xZZ = 0x00.

6.11 Single Measurement

Send: 0xAA 0x00 0x00 0x20 0x00 0x01 0x00 0x00 0x21

Receive: 0xAA 0x00 0x00 0x22 0x00 0x03 0xAABBCCDD 0x0101 Check

Slave Address: 0x00 Register Address: 0x0020/0x0022

Description: After the previous command is completed, this sets the slave to perform a single automatic measurement. The laser turns off after one measurement.

Function: Returns the measurement result to the host. Measurement Result = 0xAABBCCDD mm (Byte6=0xAA, Byte7=0xBB, Byte8=0xCC, Byte9=0xDD). Signal Quality = 0x0101. A smaller signal quality value indicates a stronger laser signal and a more reliable measurement.

6.12 Continuous Measurement

Send: 0xAA 0x00 0x00 0x20 0x00 0x01 0x00 0x04 0x25

Receive: Same feedback command as single automatic measurement.

Slave Address: 0x00 Register Address: 0x0020

Description: After the previous command is completed, this sets the slave to perform continuous automatic measurement. After Sending the command, the module will measure distance continuously at a fixed

frequency.

6.13 Slave Error Feedback

Receive: 0xEE 0x00 0x00 0x00 0x00 0x01 0xFF 0xFB 0xFC

Slave Address: 0x00 Register Address: 0x0000

Description: The slave reports an error status code to the host.

Error Status Code = 0xFFFFB (16-bit signed integer). Absolute value = 0x0005 (actual error code). Refer to section 4.7 for specific error codes.

6.14 Exit Continuous Measurement Mode

The host transmits a single byte 0x58 (ASCII character 'X') to immediately stop continuous measurement mode.

6.15 Start Multi-Slave Measurement

Send: 0xAA 0x7F 0x00 0x20 0x00 0x01 0x00 0x00 0xA0

Receive: NONE

Slave Address: 0x00 Register Address: 0x0020

Description: After the previous command is completed, this defines all slaves to be in single automatic measurement mode.

The host Sends a measurement command to the broadcast address 0x7F.

This causes all online slaves to measure distance simultaneously, but none of them will return their results to the host until the host requests each slave individually. Before Sending the read measurement result command, the host should read the status code of each slave to ensure no errors occurred during their measurement.

After Sending the broadcast command, the host polls the status of all slaves. If a slave reports a status code of 0x0000 (no error), the host can Send the read measurement result command to get the distance.

Slaves will not overwrite the previous measurement result until they successfully receive a new measurement command and measure a new distance value.

7 Status Codes

Status Code	Description	Handling Measures
0x0000	No error	–
0x0001	Input voltage is low (should be $\geq 3.5V$)	Check input/battery voltage
0x0002	Network error, can be ignored	–
0x0003	Module temperature is low ($< -20^{\circ}C$)	Increase module temperature
0x0004	Module temperature is high ($> +55^{\circ}C$)	Decrease module temperature
0x0005	Target out of range	Use within the specified measurement range in the manual
0x0006	Invalid measurement value	Re-measure
0x0007	Ambient light too strong	Reduce ambient light intensity in the measurement environment
0x0008	Laser signal too weak	Check for contamination on the output lens or increase target reflectivity
0x0009	Laser signal too strong	Decrease target reflectivity
0x000A	Hardware error 1	Contact customer support with the error code
0x000F	Laser signal unstable	Stabilize the device or check if the power supply is stable
0x0081	Invalid communication format	Check if the command was sent correctly

8 Command Reference

Function	Send Command / Received Data	Description
Check Module Status	Send: AA 80 00 00 80 Receive: AA 80 00 00 00 01 00 00 81	Read module current status Read module status: no error
Check Module HW Version	Send: AA 80 00 0A 8A Receive: AA 80 00 0A 00 01 DB 2B 91	Read module hardware version Read hardware version: 0xDB2B
Check Module SW Version	Send: AA 80 00 0C 8C Receive: AA 80 00 0C 00 01 D2 15 74	Read module software version Read software version: 0xD215
Check Module Serial No.	Send: AA 80 00 0E 8E Receive: AA 80 00 0E 00 02 F0 C8 AE 96 8C	Read module serial number version Read serial number: 0xF0C8AE96
Check Module Input V	Send: AA 80 00 06 86 Receive: AA 80 00 06 00 01 32 19 52	Read module input voltage Read input voltage: 3219 mV
Read Last Meas. Result	Send: AA 80 00 22 A2 Receive: AA 80 00 22 00 03 00 00 00 32 00 2C 03	Read last measurement result Read last result: 0x00000032 (50 mm)
Set Module Address	Send: AA 00 00 10 00 01 00 05 16 Receive: AA 00 00 10 00 01 00 05 16	Set module address (change address from 0x00 to 0x05) Module address set to 0x05 successfully
Set Module Offset	Send: AA 00 00 12 00 01 00 79 8C Receive: AA 00 00 12 00 01 00 79 8C	Set module offset to 0x79 (121 mm) Offset set successfully
Module Laser On/Off	Send: AA 00 01 BE 00 01 00 01 C1 Receive: AA 00 01 BE 00 01 00 01 C1 Send: AA 00 01 BE 00 01 00 00 C0 Receive: AA 00 01 BE 00 01 00 00 C0	Turn laser ON Laser is ON Turn laser OFF Laser is OFF
Single Measurement	Send: AA 00 00 20 00 01 00 00 21 Receive: AA 00 00 22 00 03 00 00 00 33 00 2F 87	Start single measurement Measurement data: 0x00000033 (51 mm), Signal Quality: 0x002F
Continuous Measurement	Send: AA 00 00 20 00 01 00 04 25 Receive: AA 00 00 22 00 03 00 00 00 33 00 3C 94	Start continuous measurement Measurement data: 0x00000033 (51 mm), Signal Quality: 0x003C

Appendix 2: Object Reflectivity Parameter Table

No.	Material Name	Reflectivity
1	Black Velvet	0.50%
2	Black Paper	1%~2%
3	Dark Green Leaves	1%~10%
4	Pure Water	2%
5	Black Fabric	3%
6	Asphalt	4%
7	Brown Wall	6%
8	Dark Sand & Rock	6%~10%
9	Quartz Glass	8%
10	Black Cardboard	10%
11	Earth Wall	10%
12	Black Paint	10%~15%
13	Brown Wood	12%
14	Tile	15%
15	Bright Green Leaves	15%~20%
16	Bright Sand & Rock	18%~24%
17	Clean Rough Wood Board	20%
18	Red Brick	20%
19	Young Leaves	20%~25%
20	Gray Paint	20%~30%
21	Gray Concrete	25%

22	Yellow Lacquered Fir Board	40%
23	Natural Fir Board	45%
24	White Polyethylene Plastic	60%
25	White Wall	60%
26	White Paint	60%
27	Pure White Wool Cloth	60%~70%
28	White Paper	60%~75%
29	White Enamel	70%
30	Fresh Snow Surface	70%~74%
31	White Latex Paint	80%
32	Pure White Cloth	80%~85%
33	Ground Glass Surface	85%
34	Gypsum	87%
35	White Cardboard	90%
36	Ground Silver Surface	92%
37	Kodak Standard White Board	100%